

2.Tolerances unless dimensions

Absolute Maximum Ratings (Ta=25°C)

Parameter	Symbol	Rating	Units
Reverse Voltage	V R	32	V
Operating Temperature	Topr	-20 ~ +80°C	°C
Storage Temperature	Tstg	-20 ~ +80°C	°C
Lead Soldering Temperature	Tsol	260	°C
Power Dissipation at(or below) 25°C Free Air Temperature	Pc	150	mW

Notes: *1:Soldering time $\leq$ 5 seconds.

Electro-Optical Characteristics (Ta=25°C)

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Parameter	Symbol	Condition	Min.	Typ.	Max.	Units
Rang Of Spectral Bandwidth	$\lambda_{0.5}$	$I_C=100\mu A$ $E_e=0mW/cm^2$	600	---	1200	nm
Wavelength Of Peak Sensitivity	λ_P	---	---	940	---	nm
Open-Circuit Voltage	V_{OC}	$E_e=5mW/cm^2$ $\lambda_p=940nm$	---	0.42	---	V
Rise Time	t_r	$V_R=10V$ $R_L=1000\Omega$	---	10	---	μs
Fall Time	t_f		---	10	---	
Short- Circuit Current	I_{SC}	$E_e=1mW/cm^2$ $\lambda_p=940nm$	---	3.0	---	UA
Reverse Light Current	I_L	$E_e=1mW/cm^2$ $\lambda_p=940nm$ $V_R=5V$	0.5	3.0	---	UA
Reverse Dark Current	I_D	$E_e=0mW/cm^2$ $V_R=10V$	---	10	---	na
Reverse Breakdown Voltage	BVR	$E_e=0mW/cm^2$ $I_R=100\mu A$	30	170	---	v
Total Capacitance	C_t	$E_e=0mW/cm^2$ $V_R=5V$ $f=1MHz$	---	10	---	Pf

Typical Electro-Optical Characteristics Curves

Fig.1 Collector Power Dissipation vs.
Ambient Temperature

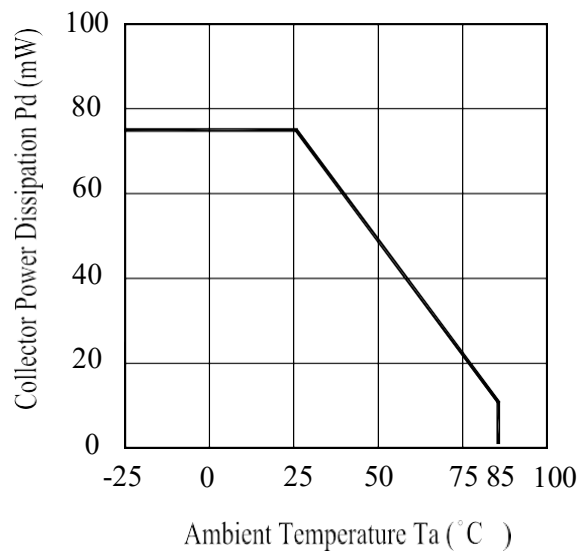


Fig.2 Spectral Sensitivity
 $T_a=25^{\circ}\text{C}$

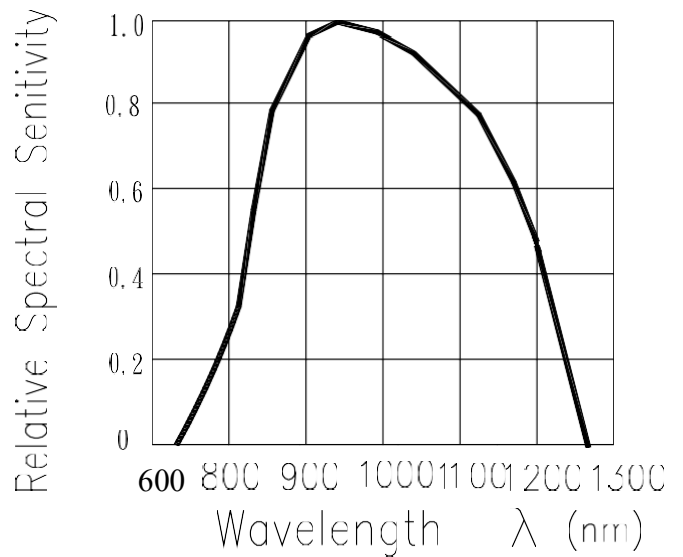


Fig.3 Relative Collector Current vs.
Ambient Temperature

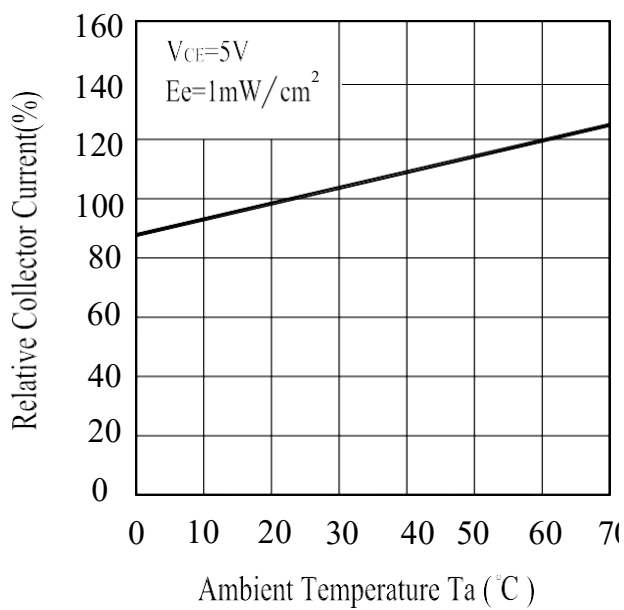
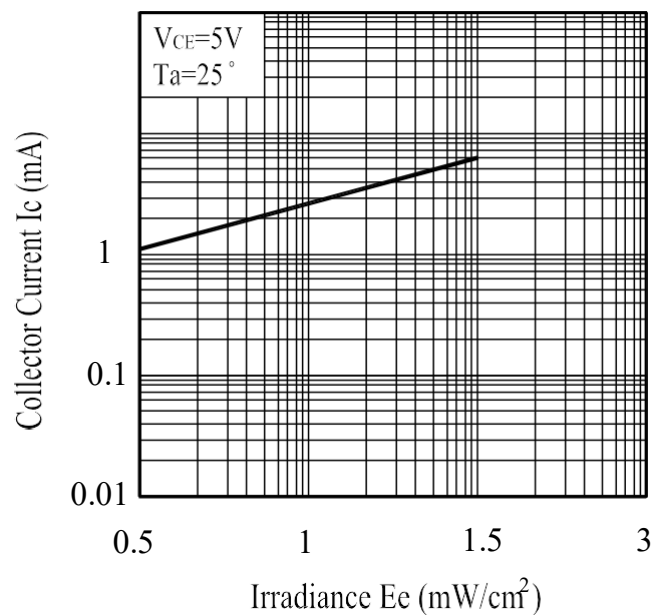


Fig.4 Collector Current vs.
Irradiance



Typical Electro-Optical Characteristics Curves

Fig.5 Collector Dark Current vs.
Ambient Temperature

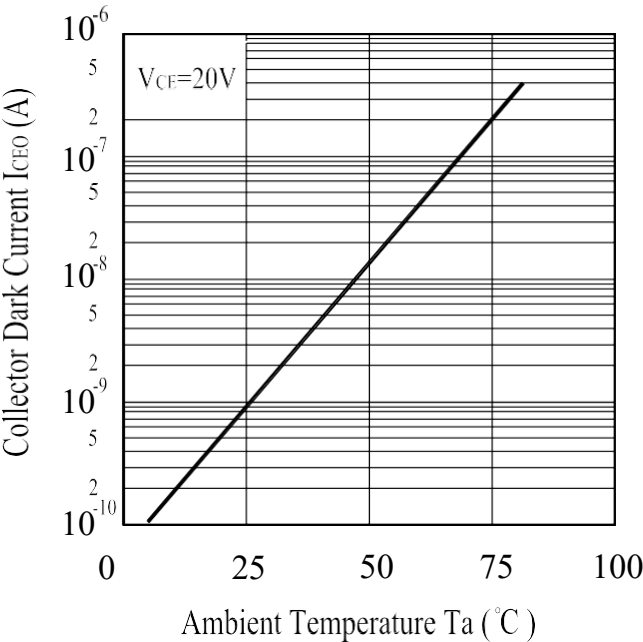
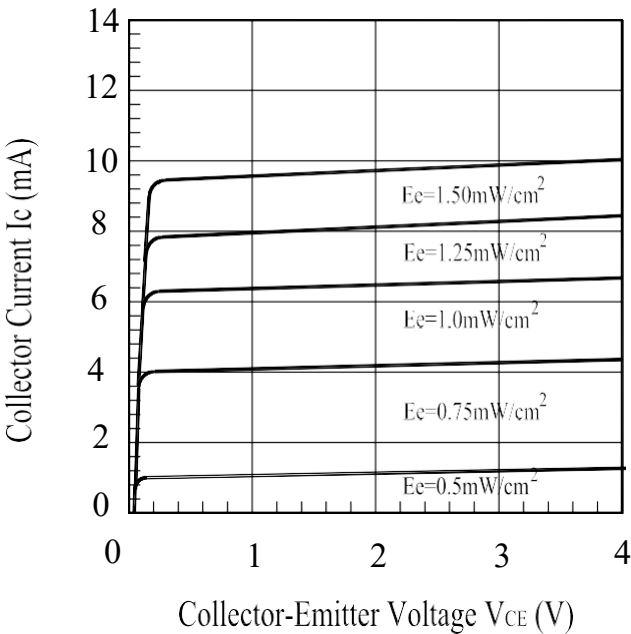


Fig.6 Collector Current vs.
Collector-Emitter Voltage



Reliability Test Item And Condition

The reliability of products shall be satisfied with items listed below.

Confidence level : 90%

LTPD : 10%

NO.	Item	Test Conditions	Test Hours/ Cycles	Sample Sizes	Failure Judgement Criteria	Ac/Re
1	Solder Heat	TEMP. : $260^{\circ}\text{C} \pm 5^{\circ}\text{C}$	10secs	22pcs	$I_{C(ON)} \leq L \times 0.8$ L : the initial test value	0/1
2	Temperature Cycle	H : $+100^{\circ}\text{C}$ 15mins ↓ 5mins L : -40°C 15mins	300Cycles	22pcs		0/1
3	Thermal Shock	H : $+100^{\circ}\text{C}$ 5mins ↓ 10secs L : -10°C 5mins	300Cycles	22pcs		0/1
4	High Temperature Storage	TEMP. : $+100^{\circ}\text{C}$	1000hrs	22pcs		0/1
5	Low Temperature Storage	TEMP. : -40°C	1000hrs	22pcs		0/1
6	DC Operating Life	$V_{CE}=5\text{V}$	1000hrs	22pcs		0/1
7	High Temperature/ High Humidity	85°C / 85% R.H	1000hrs	22pcs		0/1